

The Structure of the Møller Data Ntuple

To see the structure of the data ntuple in paw, one can type:

to see the connected units number

```
PAW> ld //
```

to see the structure of the ntuple that connected to the unit 21

```
PAW> nt/pri //lun21/1
```

Variables description:

```
*****
* Var numb * Type * Range * Block * Name * Description
*****
* 1 * U*4 * [1,16000] * RUN * irun - run number
* 2 * U*4 * [0,255] * RUN * idtyp - data type
* 3 * U*4 * [0,255] * RUN * ievtyp - event type
* 4 * I*4 * * * RUN * iret - debug/errors code
* 5 * U*4 * [0,1] * RUN * itrig(8) - VME TI board input regi
* 6 * I*4 * * * RUN * itick - ticks number between ev
* 1 * I*4 * [0,36] * ADC * nadc - number of adc channels
* 2 * U*4 * [0,4095] * ADC * iadc(nadc) - adc channels data
* 1 * I*4 * [0,4096] * TDC * ntdc - number of tdc hits
* 2 * U*4 * [0,256] * TDC * itch(ntdc) - tdc channel number with
* 3 * U*4 * [0,60000] * TDC * itim(ntdc) - tdc hits time (resoluti
* 4 * U*4 * [0,1] * TDC * ited(ntdc) - tdc type of hits
* 5 * I*4 * [0,10] * TDC * nhcha -
* 6 * U*4 * [0,1000] * TDC * ihit(nhcha) -
* 1 * I*4 * [0,33] * SCAL * nsca - number of scalers
* 2 * I*4 * * * SCAL * isca(nsca) - scalers data
*****
```

The map of ADC, TDC, Scaler and TI signals to corresponded channels:

```
# Last update 17 Jun 2019
# 2005 - encoders
# Cosmetic update 22 Apr 2008
# Update with BPMs 17 Jun 2019
#
# Definitions:
# 30 Hz helicity flip rate:
#     MPS - signal 250us long at the helicity change period, inverted
#     QRT - signal 33 ms long for the 1-st window in a train (quadruplet)
#
#
#
Discriminator Module OrtecTD8000 CAMAC slot 1
inputs/outputs signal
1 SumLeftLed
2 SumRightLed
3 SumLeftRight
4 ApertLeft
```

5 ApertRight
 6 -
 7 Sum from the 16xdiscriminator (8ns)

Delay Module Lecroy4518 CAMAC slot 6

inputs/outputs	signal
1	SumLeftLed
2	SumRightLed
3	SumLeftRight
4	ApertLeft
5	ApertRight
6	SumLeftDelayed (114ns cable delay installed 5 Apr 1998)
7	ApertLeftDelayed (114ns cable delay installed 5 Apr 1998)
8	Pulser
9	LED pulser
10	-
11	Sum from 16 x discrim (TD80000 out 7)
12	input: Trigger (from PLU out 8 0.5ns) output: -> ADC-2 gate

PLU Module Lecroy2365 CAMAC slot 11

inputs	signal
1	SumLeftLed
2	SumRightLed
3	SumLeftRight
4	ApertLeft
5	ApertRight
6	BCM
7	DelaySumLeftLed2
8	- faulty input (was DelayApertLeft2 before 4 Apr 99)
9	Sum from 16 x discrim (Delay out 11)
10	!MPS from the counting house removed on 18/09/2002
11	Clock (100kHz) TimerGenerator
12	DelayApertLeft2 (from 8/4/99)
13	MPS
14	TIR output 3 (starting with 1)
15	Pulser (used for the pulser mode) - 80ns - overlaps inp 1,2,..
16	a pulse for synchronisation with the LED (from the LED pulser)

outputs	signal
1	SingleLeft * input(13)
2	SingleRight * input(13)
3	Coinc * input(13)
4	Accidental coincidence* input(13)
5	BCM * input(13)
6	TimerGenerator * input(13)
7	- (connected to scaler 6) detsetup : LeftLG*LeftAppDelayed*input(13) LED setup : input(16)*input(13)
8	AdcGate * input(13) (gate width = 90ns)

TDC 2277

inputs: first 16 = delay unit outputs

1	SumLeftLed
2	SumRightLed
3	SumLeftRight
4	ApertLeft
5	ApertRight
6	SumLeftDelayed (114ns cable delay installed 5 Apr 1998)
7	ApertLeftDelayed (114ns cable delay installed 5 Apr 1998)
8	the pulser signal
9	LED pulser

```

11      Sum 16 x discr
12      Trigger ADC-2 gate

18      Coincidence * PLUinput(13) * "delayed" helicity

```

Scaler 1 Module LeCroy 1151E

```

inputs      signal
1           SingleLeftPl
2           SingleRightPl
3           CoincPl
4           AccidentalPl
5           BeamChargePl
6           PLU out 7; in LED setup - number of LED flashes
7
8
9
10          (Helicity delayed)*Clock(100kHz) (from 22 jul 2003, run 10952)
11          !MPS
12          PLU out #6 (TimerGenarator * helicity window)
15          Clock TimerGenerator

```

Scaler 2 Module LeCroy 1151E

```

inputs      signal
1           SingleLeftPl
2           SingleRightPl
3           CoincPl
4           AccidentalPl
5           SumLeftLG delayed (from 08/09/99)
6           SumRightLG
7           SumLG
8           AppLeft    delayed
9           AppRight
10          +X- BPM 1H01 from VtoF convertor (since 06/13/2019) # before: Encc
11          +X+ BPM 1H01 from VtoF convertor (since 06/13/2019) # before: Encc
12          +Y+ BPM 1H01 from VtoF convertor (since 06/13/2019)
13          +Y- BPM 1H01 from VtoF convertor (since 06/13/2019)

```

VME Trigger Interface

```

1  trigger 0  ADC gate
2           1  (!MPS) * live_time 100ns
3           2
4           3
5  input  A   -
6         B   Helicity delayed
7         1   live time
8         2   QRT (from 18/09/2002)

      output
1         0   =1 - send a signal to reset the dead time (should be 1,0)
2         1   =0/1 - pulser off/on
3         2   - to input 14 of the PLU : disable trigger_1
6         5   =_|-|_ - strobe pulse(CLK) for latch DATA, loading to LED gen.
7         6   =1/0 - disable/enable the pulser for LED (serial DATA output) -> E
8         7   =0/1 - scaler gate off/on

```